The earthquake

1. What is an earthquake?

An earthquake is a natural phenomenon in which the Earth's crust moves suddenly, as a result of the release of a large amount of energy that has built up inside the Earth.

This energy is transmitted in the form of seismic waves, which spread in all directions and cause vibrations of the ground. Earthquakes can last from a few seconds to a few minutes, and their intensity can vary from very weak to extremely powerful.

Earthquakes are among the most dangerous natural phenomena, because they occur suddenly and can cause significant damage.







2. The causes of earthquakes

The main cause of earthquakes is the movement of the tectonic plates. The Earth's crust is divided into several huge plates that move slowly. When these plates collide, move apart or slide past one another, the accumulated energy is released in the form of an earthquake.

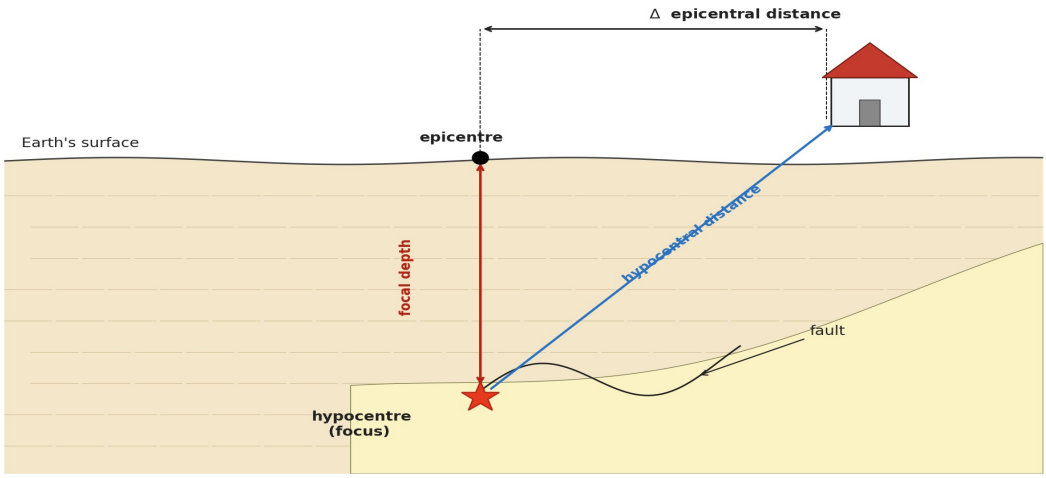
There are several types of causes:

- **Movement of tectonic plates** – the most common and important cause
- **Volcanic activity** – earthquakes occur in areas with active volcanoes
- **Underground collapses** – in caves or mines
- **Human activities** – explosions, resource extraction

3. The elements of an earthquake

The **hypocentre** (focus) is the place inside the Earth where the initial rupture occurs and from which the earthquake starts.

The **epicentre** is the point on the surface of the Earth located directly above the hypocentre. This is where the strongest effects of the earthquake are felt.



Cross-section showing the hypocentre, the epicentre, the focal depth, the epicentral distance, the hypocentral distance and the fault.

Seismic waves are the vibrations that carry the energy from the focus toward the surface. There are several types of waves:

- **primary waves (P)** – fast and less destructive
- **secondary waves (S)** – slower, but more dangerous
- **surface waves** – the most destructive

4. Measuring earthquakes

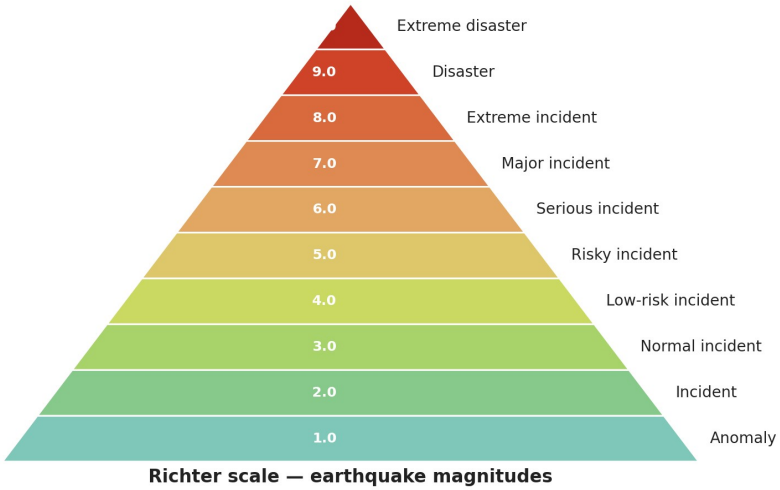
Earthquakes are analysed and measured with the help of special instruments called seismographs. These record the movements of the ground and help with determining the intensity and the magnitude.

The **magnitude** represents the amount of energy released during the earthquake. It is measured on the Richter scale or on the moment-magnitude scale.

The **intensity** represents the effects produced on people, buildings and the environment. It is measured on the Mercalli scale.

General classification:

- **under 3** – very weak, barely perceptible
- **3–5** – weak to moderate
- **5–7** – strong
- **over 7** – very strong and dangerous



Visual representation of earthquake magnitudes on the Richter scale, from minor anomalies to extreme disasters.

5. Effects and protective measures

Earthquakes can have serious consequences:

- destruction of buildings and infrastructure
- loss of human lives
- fires and explosions
- landslides
- tsunamis (in the case of underwater earthquakes)

The impact depends on the intensity, the depth and the level of preparedness of the population.

In order to reduce the risks, it is important to know how to act:

During the earthquake:

- stay calm and do not panic
- take shelter under a sturdy table

- stay away from windows and from objects that can fall
- do not use the lift

After the earthquake:

- leave the building carefully
- check whether there are any injured people
- avoid dangerous areas
- follow the instructions of the authorities